

Deep Learning for teens

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Quantizing in DL

- ▶ Used to make big model the smaller one
- ▶ Then u can deploy them on edge devices like micro controller
- ▶ There two way to perform quantization in tensorflow
 - ▶ Post training quantization
 - ▶ Quantization aware training

Post training quantization

- ▶ U take the model and use tf.lite convert to make the model smaller in size
 - ▶ This is a quick approach
 - ▶ Down side is accuracy might get suffered . Better approach is using
 - ▶ Quantization aware training

Quantization aware training

- ▶ You take the model, then u apply quantized function on it, then u take quantized model in tensorflow
- ▶ Then u do training again (like transfer learning) for fewer epochs
- ▶ U get fine tuned quantized model
- ▶ Then u use tf lite (if u want to deploy models on edge devices , u need to use tf lite)

Assignment

- ▶ Use the *MNIST* dataset and repeat the exact same steps we followed.